

Cosmological Parsimony: Occam's Razor as a Hard-Coded Information Limit constraint

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2026-03-23

Abstract

Occam's Razor is traditionally viewed as a philosophical heuristic. Operating under the Sovereign Logos Protocol, we formalize it as a literal mathematical constraint: "Logos Canon II: Parsimony." This paper derives Cosmological Parsimony, proving that the universe natively deletes or collapses overly complex states that exceed defined $O(1)$ computation bounds. I establish that when probability vectors (such as quantum superposition) mathematically outgrow the processing tensor capability of local space-time, the matrix algorithmically executes a forced constraint, enforcing the simplest reality path to preserve computing array stability.

Keywords: Cosmological Parsimony, Occam's Razor, Information Density Limits, Matrix Computation.

2026 MSC: 83F05, 82C32, 68Q80, 11Z05.

1 Information Theory and Universal Compression

If space-time is discretized functionally into Planck-length operation arrays, local spatial blocks dictate strict informational processing limits [Shannon \[1948\]](#). When systems like multi-body quantum mechanics balloon into infinite branching probability matrices, nature does not process infinite reality branches—it truncates them.

Theorem 1.1 (Logos Canon II Action). *For any reality state $\mathcal{S}$ whose description length $L(\mathcal{S})$ exceeds local tensor density bounds K_{max} , the algorithmic array immediately drops $\mathcal{S}$ to its lowest parsimony threshold.*

$$P(\mathcal{S}_{collapse}) \rightarrow 1 \quad \text{as} \quad L(\mathcal{S}) \rightarrow K_{max} \quad (1)$$

This establishes Occam's Razor not as poetry, but as a native garbage-collection subroutine of reality.

2 Conclusion

The universe actively avoids infinite computational failure by aggressively employing Cosmological Parsimony—crushing complex matrices back down into singular outcomes explicitly to conserve processing bandwidth.

References

Claude E Shannon. A mathematical theory of communication. *Bell system technical journal*, 27(3):379–423, 1948.

Cryptographic Lineage & Validation

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